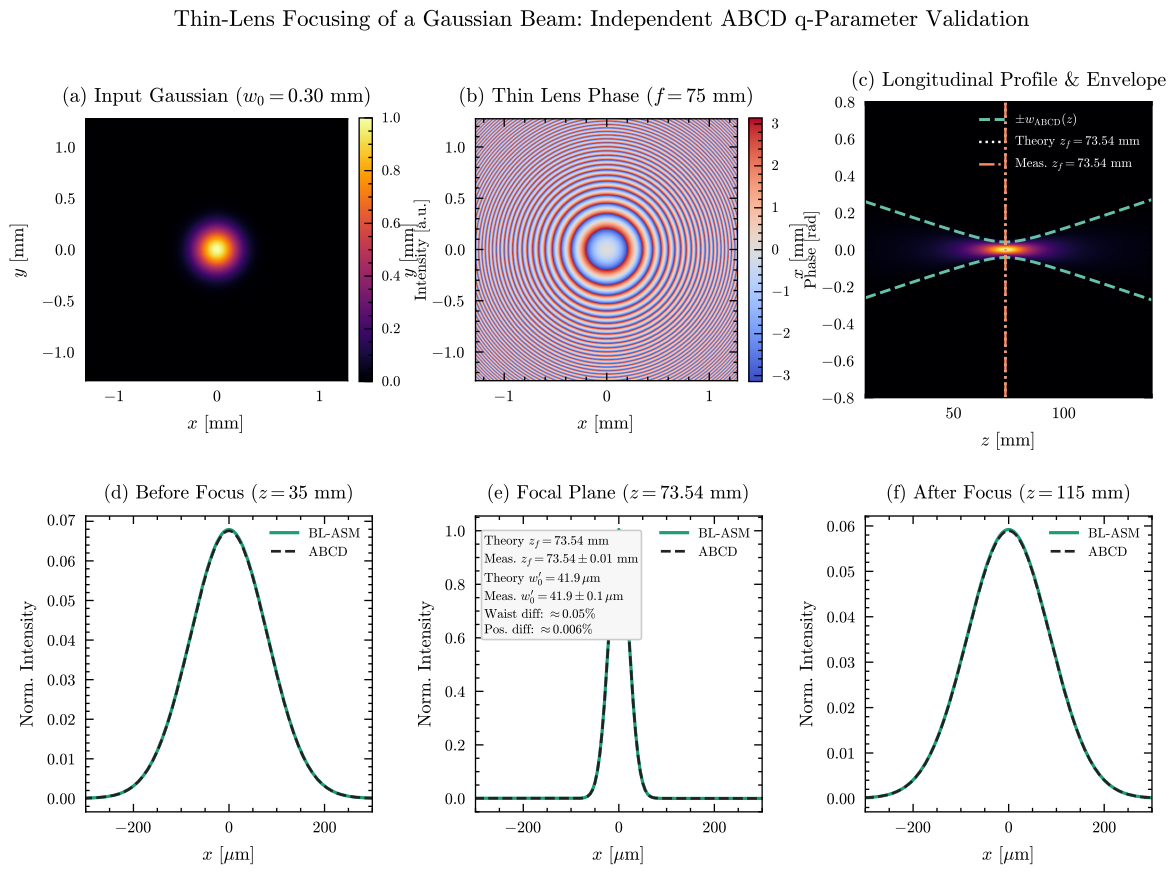

Thin-Lens Focusing of a Gaussian Beam: Independent ABCD q-Parameter Validation
(a) Input Gaussian ($w_0 = 0.30$ mm)
(b) Thin Lens Phase ($f = 75$ mm)
(c) Longitudinal Profile & Envelope
$\pm w_{\mathrm{ABCD}}(z)$
Theory $z_f = 73.54$ mm
Meas. $z_f = 73.54$ mm
x [mm]
y [mm]
Intensity [a.u.]
Phase [rad]
z [mm]
(d) Before Focus ($z = 35$ mm)
(e) Focal Plane ($z = 73.54$ mm)
(f) After Focus ($z = 115$ mm)
BL-ASM
ABCD
Norm. Intensity
x [μm]
Theory $z_f = 73.54$ mm
Meas. $z_f = 73.54 \pm 0.01$ mm
Theory $w_0' = 41.9\,\mu$m
Meas. $w_0' = 41.9 \pm 0.1\,\mu$m
Waist diff: $\approx 0.05\%$
Pos. diff: $\approx 0.006\%$